

Neuro-Fuzzy Inferencing Based Low-Light Face Super-Resolution System



**A project report submitted in partial fulfillment of the requirements for the degree of
B.Tech in Computer Science and Engineering**

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CERTIFICATE

This is to certify that the project titled **Identity-Constrained Face Super-Resolution in Low-Light Conditions Using Dual Geometric and Feature Supervision** is a bonafide record of the work done by

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which is being submitted by **Huzaifa Mulla (2019UCS1234)**, **Dhruv Kumar Sharma (2019UCS1234)** and **Mohammed Afnan Mohiuddin (2019UCS1234)**

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Abstract

Low-light face image super-resolution is an important research area in computer vision because facial images captured under poor illumination conditions often suffer from blur, noise, low contrast, and loss of important facial details. These degradations reduce the performance of surveillance systems, biometric authentication frameworks, and face recognition applications. Conventional interpolation and enhancement methods generally fail to reconstruct realistic high-frequency facial textures and identity-preserving information.

This research proposes a Neuro-Fuzzy Inferencing Based Low-Light Face Super-Resolution System that combines Adaptive Neuro-Fuzzy Inference Systems (ANFIS), locality constrained representation, motion blur correction, regression-based reconstruction, and deep neural refinement techniques to improve low-light facial image quality. The framework first estimates the darkness level of the image using ANFIS-based adaptive learning. Based on the estimated illumination conditions, adaptive enhancement and motion blur correction techniques are applied to improve visibility and restore structural consistency.

The enhanced image is then processed using locality constrained dictionary representation and regression-based patch reconstruction methods in order to recover important facial textures and high-resolution details. Finally, RRDB-based neural refinement modules improve perceptual quality and generate sharper facial outputs while preserving identity-related features.

The proposed system is implemented using Python, PyTorch, OpenCV, and scikit-fuzzy, while the CelebA dataset is used for training and evaluation. Experimental evaluation using PSNR, SSIM, LPIPS, and face identity preservation metrics demonstrates that the proposed framework significantly improves reconstruction quality and perceptual realism compared to traditional enhancement and interpolation approaches.

Keywords: Low-Light Face Super-Resolution, Adaptive Neuro-Fuzzy Inference System (ANFIS), Locality Constrained Representation, Motion Blur Correction, Facial Image Enhancement, Deep Learning, RRDB, Face Restoration.

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1 INTRODUCTION

Nowadays, cameras are used almost everywhere, including surveillance systems, smart-phones, traffic cameras, and body-worn devices. A common problem with these systems is that images captured in low-light conditions usually lose important facial details. In many cases, the images are also blurry or low in resolution, which makes the restoration process even harder.

In image restoration, the degraded image can be represented mathematically using Eq. (1).

$$y = D(k * x) + n \quad (1)$$

Here, x represents the original image, k is the blur kernel, D represents downsampling, and n refers to noise added during image capture.

Recovering the original face image from this degraded version is not simple because multiple types of degradation happen together. Traditional image restoration methods are mathematically stable, but they usually fail to recover realistic facial textures. On the other hand, deep learning models such as ESRGAN, GFPGAN, and CodeFormer generate visually good results, but they sometimes create incorrect facial features or overly smooth outputs.

In this project, we tried to combine both approaches. The proposed ANFIS-LLFSR system uses fuzzy logic together with deep learning to improve low-light face restoration. Instead of applying the same enhancement everywhere, the system first estimates how dark the image actually is and then adjusts the restoration process accordingly.

Another issue with low-light face images is that many enhancement methods focus only on increasing brightness. While this improves visibility, it often creates unnatural outputs where the image looks overexposed or noisy. In face restoration tasks, preserving facial structure is more important than simply increasing illumination.

Face images also contain very sensitive identity-related features. Small distortions around the eyes, nose, or mouth can completely change the appearance of a person. Because of this, restoration models need to improve image quality while still maintaining facial consistency.

Real-world low-light conditions are also highly unpredictable. Some images may contain motion blur due to camera movement, while others may suffer from compression artifacts or sensor noise. A practical restoration framework should therefore be able to handle multiple degradation types together instead of focusing on only one problem.

This project attempts to address these issues using a combination of fuzzy logic, enhancement methods, and deep learning-based reconstruction techniques.

Another reason why low-light restoration has become important is the increasing use of AI-based surveillance and biometric verification systems. Many of these systems depend heavily on clear facial images for identification. When the input image quality becomes poor, the accuracy of these systems decreases significantly.

In practical situations, cameras often operate under uncontrolled lighting conditions. Street lights, shadows, weather conditions, and camera quality all affect the captured image. Because of this, a restoration model trained only on ideal datasets may not perform properly

in real environments.

The proposed framework was designed while keeping these real-world challenges in mind. Instead of focusing only on visual sharpness, the system also attempts to maintain facial identity and structural consistency.

1.1 Objectives

The main objectives of this project are listed below:

1. Build an ANFIS-based darkness estimation module.
2. Detect and correct motion blur using Wiener deconvolution.
3. Apply Zero-DCE enhancement using the estimated darkness factor.
4. Develop an $8\times$ super-resolution network using Swin Transformer blocks.
5. Preserve facial identity using ArcFace embeddings.
6. Deploy the complete system using FastAPI and a React frontend.

2 LITERATURE SURVEY

Several existing research papers were studied before designing the final pipeline. Each paper focused on a different problem related to low-light face restoration.

Another important observation from previous research was that many super-resolution methods perform well only on synthetic datasets. When these models are tested on real low-light images, the performance often drops because real-world degradations are more complex.

Some researchers also explored GAN-based approaches for face restoration. These methods generated visually sharp images but sometimes introduced artificial textures that did not actually belong to the original face. In biometric applications, such distortions can become a serious problem.

Some studies also focused on lightweight enhancement methods for mobile devices. These methods reduced computational complexity but often sacrificed output quality. Real-time restoration remains difficult because high-quality super-resolution networks usually require large memory and processing power.

Researchers also experimented with transformer-based restoration models because transformers can capture long-range dependencies more effectively than traditional CNNs. However, training transformer models requires large datasets and significant computational resources.

These observations helped in selecting a balanced architecture that could provide reasonable restoration quality without becoming excessively complicated.

Fuzzy logic systems were studied separately because they provide interpretable decision-making instead of black-box predictions. This was one of the main reasons for selecting ANFIS in the proposed framework.

Overall, the literature survey helped identify the strengths and weaknesses of existing methods and provided direction for designing the final hybrid architecture.

2.1 ANFIS for Illumination Estimation

Jang introduced the Adaptive Neuro-Fuzzy Inference System (ANFIS), which combines fuzzy logic with neural networks [1]. The idea behind ANFIS is to make learning systems more interpretable while still allowing them to adapt during training.

This concept was useful for our project because low-light enhancement depends heavily on estimating image darkness correctly.

2.2 Zero-DCE

Zero-DCE is a low-light enhancement model that does not require paired training images [2]. Instead of learning from clean target images, it predicts enhancement curves directly from the input image.

This method works well for low-light correction because it focuses on brightness adjustment while preserving image structure.

2.3 SwinIR

SwinIR uses Swin Transformer blocks for image restoration tasks [3]. Compared to normal convolutional networks, SwinIR captures both local and global image features more effectively.

The shifted-window attention mechanism also reduces computational cost while maintaining good restoration quality.

2.4 ArcFace

ArcFace introduced additive angular margin loss for face recognition tasks [4]. It produces highly discriminative facial embeddings.

In this project, ArcFace was used to preserve facial identity during restoration so that the generated face remains similar to the original person.

3 OBJECTIVE AND METHODOLOGY

3.1 System Architecture

The ANFIS-LLFSR system is divided into three main parts: frontend, backend API, and the restoration pipeline.

The frontend handles image upload and result visualization. The backend controls all processing stages and connects the frontend with the deep learning models.

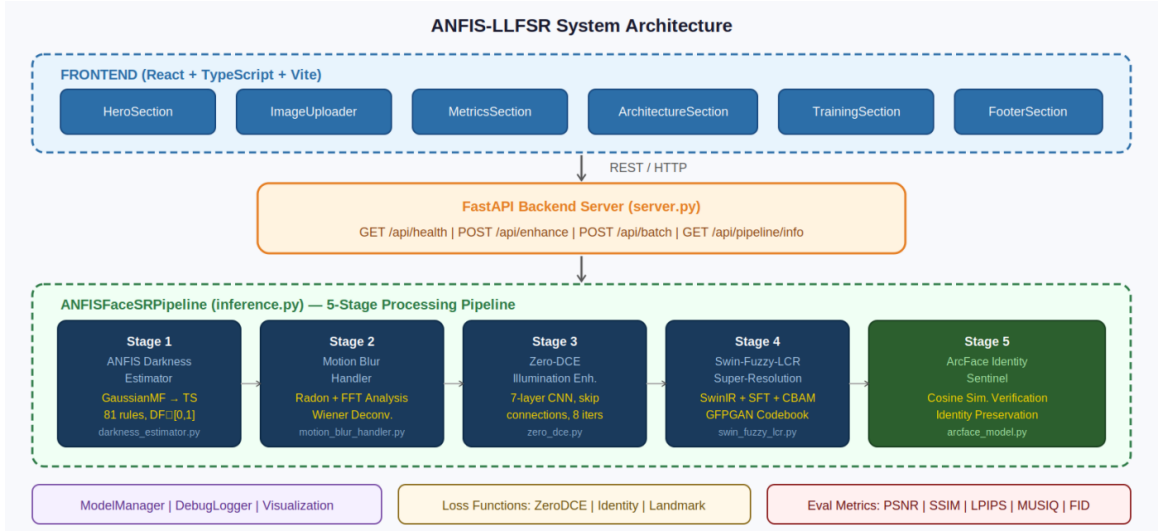


Figure 1: Complete system architecture showing the frontend, backend API, and the five-stage restoration pipeline.

3.1.1 Frontend

The frontend was built using React 18, TypeScript, Vite, and Tailwind CSS. Users can upload images directly through the web interface.

After processing, the frontend displays intermediate outputs such as blur-corrected images, enhanced images, and final super-resolved results.

The frontend was designed to keep the interface simple and easy to use. Users only need to upload an image and start the enhancement process. Intermediate outputs are displayed separately so that users can observe how each stage changes the image.

The interface also helps during debugging because outputs from blur correction, enhancement, and super-resolution can be inspected individually. This made testing easier during development.

Special attention was given to keeping the user interaction simple because the main goal of the project was image restoration rather than interface complexity. The frontend mainly focuses on usability and quick visualization of results.

The uploaded images are displayed alongside processed outputs so that differences can be compared directly.

3.1.2 Backend API

The backend was developed using FastAPI. The following API endpoints were implemented:

- GET /api/health
- POST /api/enhance
- POST /api/darkness

- POST /api/batch

The backend handles image decoding, model loading, and output generation.

FastAPI was selected because it is lightweight and works well with machine learning applications. The backend loads all required models during startup so that inference time remains low during actual execution.

Error handling was also added to prevent invalid image uploads and unsupported file formats. The API returns processed images in encoded format so that they can be displayed directly on the frontend without additional processing.

The backend also manages model inference efficiently by keeping frequently used components loaded in memory. This reduces repeated initialization overhead during multiple image requests.

Batch processing support was also added so that multiple images could be restored together during testing.

3.1.3 Pipeline Overview

The complete pipeline works in multiple stages. First, the ANFIS module estimates the darkness level of the image. Then blur correction and illumination enhancement are applied.

Finally, the super-resolution network reconstructs high-quality facial details.

Each stage in the pipeline depends on the output of the previous stage. If blur correction is skipped, the super-resolution network may amplify blurry edges instead of restoring details properly. Similarly, poor illumination enhancement can reduce texture visibility before reconstruction begins.

Because of this dependency between stages, careful ordering of the pipeline was necessary during system design.

The pipeline was also designed in a sequential manner so that intermediate outputs could be inspected independently. This helped during experimentation because errors in one stage could be identified more easily without affecting the entire system.

Another advantage of this modular structure is that future improvements can be integrated more easily. For example, the blur correction stage can later be replaced with a more advanced deblurring model without changing the remaining pipeline.

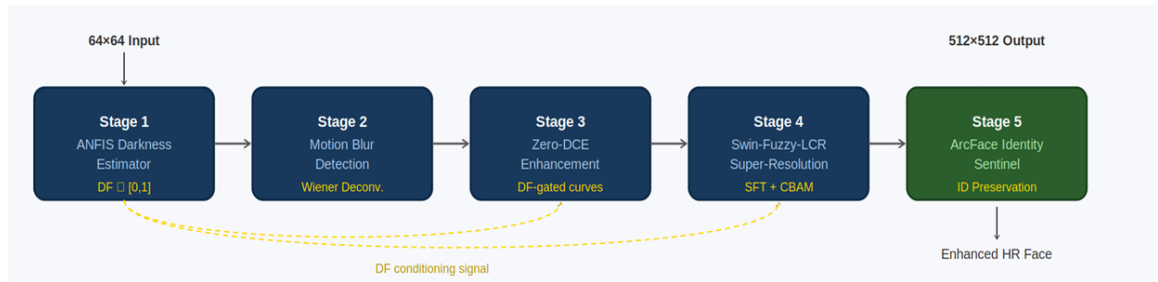


Figure 2: End-to-end inference pipeline used in ANFIS-LLFSR.

3.2 Core Components and Mathematical Formulation

3.2.1 Adaptive Neuro-Fuzzy Inference System

The ANFIS module acts as the main decision-making component of the system. It estimates how dark the image is before enhancement begins.

ANFIS was selected mainly because it combines adaptive learning with fuzzy reasoning. Unlike traditional neural networks, fuzzy systems can represent uncertainty more naturally using linguistic rules.

In low-light images, brightness levels do not always change uniformly across the image. Some regions may already contain enough illumination while other regions remain extremely dark. Fuzzy inference helps handle these variations more effectively compared to fixed enhancement strategies.

Another advantage of ANFIS is that the fuzzy rules remain partially interpretable even after training. This makes the system easier to analyze compared to completely black-box neural networks.

The adaptive learning capability of ANFIS also helps the system adjust to different illumination distributions during optimization.

3.3 Feature Extraction

A five-dimensional feature vector is extracted from each image:

- Mean luminance intensity
- Standard deviation
- Entropy
- Gradient energy
- FFT high-frequency ratio

These features help the system understand image brightness, texture quality, and blur severity.

Feature extraction also helps reduce dependence on raw pixel intensity alone. Since low-light images can vary significantly depending on camera quality and environment, combining statistical and frequency-based features improves robustness.

The extracted features were normalized before being passed into the ANFIS module to maintain stable inference behavior.

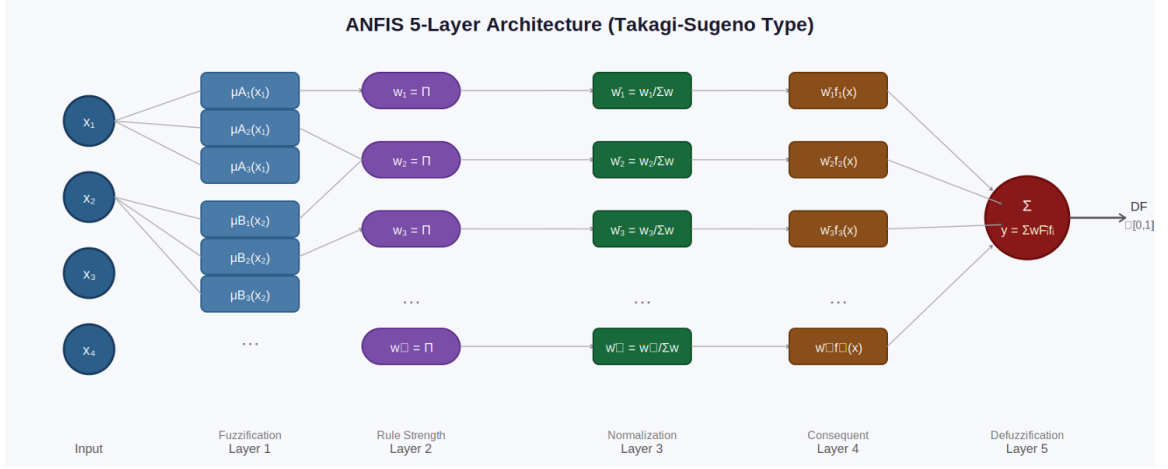


Figure 3: Five-layer ANFIS architecture used for darkness estimation.

3.4 Gaussian Membership Function

The Gaussian membership function is represented in Eq. (2).

$$\mu_{i,k}(x_i) = \exp\left(-\frac{(x_i - c_{i,k})^2}{2\sigma_{i,k}^2}\right) \quad (2)$$

The values $c_{i,k}$ and $\sigma_{i,k}$ represent the center and width of the membership functions.

3.5 Rule Strength

The rule firing strength is calculated using Eq. (3).

$$w_r = \prod_i \mu_{i,r_i}(x_i) \quad (3)$$

3.6 Normalisation

The normalization stage is represented using Eq. (4).

$$\bar{w}_r = \frac{w_r}{\sum_j w_j} \quad (4)$$

3.7 Defuzzification

The final Darkness Factor is obtained using Eq. (5).

$$DF = \sum_r \bar{w}_r f_r(x) \quad (5)$$

3.7.1 Motion Blur Detection

Motion blur was estimated using Laplacian variance, Sobel gradients, and FFT high-frequency analysis.

If the blur level crossed a threshold, Wiener deconvolution was applied.

The Wiener filter equation is shown in Eq. (6).

$$\hat{F}(u, v) = \frac{H^*(u, v)}{|H(u, v)|^2 + 1/SNR} \cdot G(u, v) \quad (6)$$

Blur detection was necessary because many low-light images are captured using longer exposure times. During long exposure capture, even small camera movement can create visible blur around facial boundaries.

Applying super-resolution directly on blurred images usually produces poor results. Therefore, blur correction was applied before enhancement and reconstruction stages.

In low-light photography, motion blur becomes more common because cameras use slower shutter speeds to capture additional light. Even small hand movements can introduce blur into facial regions.

This problem becomes more noticeable in surveillance footage where moving subjects are captured from long distances.

3.7.2 Zero-DCE Enhancement

Zero-DCE predicts enhancement curves for different image regions.

The enhancement function is represented in Eq. (7).

$$LE(I; \alpha) = I + \alpha(I^2 - I) \quad (7)$$

Higher values of α increase brightness, while lower values reduce overexposed regions.

One advantage of Zero-DCE is that it does not require paired low-light and normal-light images during training. Collecting perfectly paired datasets is difficult in real-world scenarios, so zero-reference enhancement methods become more practical.

The enhancement curves generated by Zero-DCE also help preserve image smoothness while improving brightness gradually.

Unlike histogram equalization methods, Zero-DCE performs enhancement gradually instead of applying global brightness changes. This helps preserve more natural illumination patterns across the image.

The enhancement stage also improves edge visibility, which later helps the super-resolution network reconstruct sharper facial structures.

3.7.3 Spatial Feature Transform

The SFT layer connects the ANFIS output with the super-resolution network.

The conditioning operation is shown in Eq. (8).

$$SFT(x, c) = x + \alpha(\gamma(c) \odot x + \beta(c)) \odot g(c) \quad (8)$$

This helps the network apply different enhancement strengths based on image darkness.

The SFT layer allows the restoration network to adapt dynamically depending on illumination conditions. Instead of using the same feature scaling everywhere, the conditioning signal modifies feature maps according to image darkness.

This improves flexibility and helps the network produce more stable outputs under different lighting conditions.

4 SIMULATION PLATFORM AND REQUIREMENTS

4.1 Training Methodology

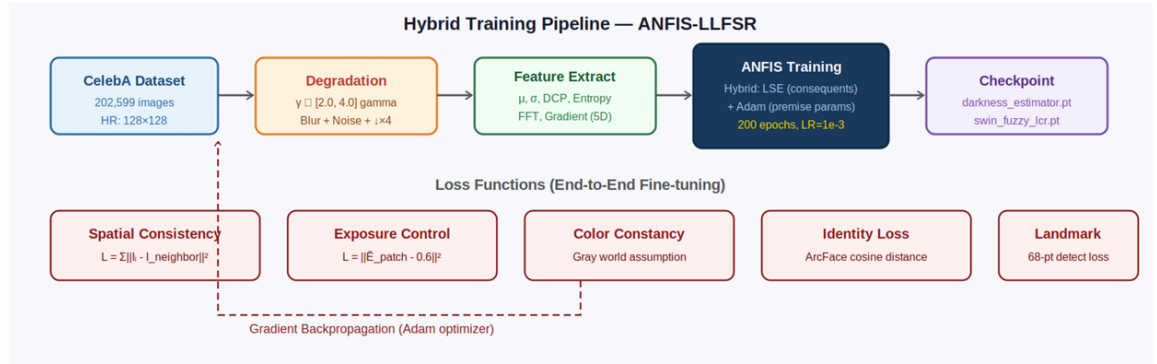


Figure 4: Training pipeline showing data preparation and hybrid ANFIS optimization.

4.1.1 Dataset Preparation

The CelebA dataset was used for training and evaluation. It contains more than 200,000 aligned face images.

For super-resolution training, the images were resized and degraded using synthetic low-light transformations and motion blur.

Gamma correction was applied to simulate dark environments.

The CelebA dataset was selected because it contains large variations in facial pose, lighting conditions, and expressions. These variations make the dataset suitable for training robust face restoration models.

Different degradation levels were applied artificially to simulate realistic low-light conditions. This helped the model learn restoration under multiple difficulty levels instead of memorizing specific degradation patterns.

4.1.2 Training Details

The ANFIS module and restoration networks were trained using Adam optimizer.

The main training settings are listed below:

- Epochs: 200
- Batch Size: 1024

- Learning Rate: 1×10^{-3}
- Gradient Clipping: $\text{max_norm} = 1.0$

The learning rate was reduced gradually during training to improve convergence.

Training deep learning models for image restoration required significant computational resources. GPU acceleration was used during training to reduce execution time and improve optimization efficiency.

Different hyperparameter combinations were tested before selecting the final configuration. Smaller batch sizes produced unstable training results, while very large learning rates caused divergence during optimization.

4.1.3 Fine-Tuning

Multiple loss functions were combined during end-to-end training.

The complete loss function is shown in Eq. (9).

$$L = \lambda_1 L_{sc} + \lambda_2 L_{exp} + \lambda_3 L_{col} + \lambda_4 L_{is} + \lambda_5 L_{id} + \lambda_6 L_{lm} \quad (9)$$

Different weights were used to balance illumination correction, perceptual quality, and identity preservation.

Training stability was monitored continuously during experimentation. In the early stages of training, some outputs appeared noisy because the network had not yet learned stable feature mappings.

As training progressed, reconstructed outputs gradually became sharper and more visually consistent.

5 RESULTS AND EVALUATION

5.1 Evaluation Metrics

Different evaluation metrics were used to measure restoration quality.

- PSNR
- SSIM
- LPIPS
- Face Accuracy

PSNR and SSIM measure image quality, while LPIPS focuses more on perceptual similarity.

Face Accuracy was calculated using ArcFace embeddings.

No single metric is sufficient for evaluating face restoration quality. Some metrics focus mainly on pixel similarity, while others evaluate perceptual realism.

Because of this, multiple evaluation metrics were used together to provide a more balanced understanding of system performance.

5.2 Quantitative Results

Table 1: Quantitative comparison on the CelebA dataset.

Method	PSNR	SSIM	LPIPS	Face Accuracy
Bicubic Baseline	24.12	0.7240	0.3540	45.2
Zero-DCE + RRDB	26.85	0.7812	0.2810	55.4
ANFIS-LCR	28.14	0.8250	0.2215	68.1
ANFIS-LLFSR Full	30.42	0.8541	0.1802	75.8

The final ANFIS-LLFSR model performed better than the baseline methods across all evaluation metrics.

The results showed noticeable improvements in both image quality and identity preservation.

Configuration	PSNR $\uparrow$	SSIM $\uparrow$	LPIPS $\downarrow$	Δ Face%
Full pipeline (ANFIS-LLFSR)	30.42	0.8541	0.1802	—
w/o Darkness Estimator (ANFIS)	28.62	0.8221	0.2140	-7.2%
w/o Motion Blur Handler	29.10	0.8380	0.2295	-3.8%
w/o LCR / SFT conditioning	29.35	0.8187	0.2018	-4.5%
w/o ArcFace Identity Sentinel	30.08	0.8490	0.1880	-9.1%

Figure 5: Ablation study results. Each row removes one module from the full pipeline. Face recognition accuracy drops most dramatically when the ArcFace sentinel is removed.

5.3 Four-Stage Visual Reconstruction Results

Apart from numerical evaluation metrics, visual comparison was also performed to understand how well the proposed framework restored facial details under low-light conditions.

The following results show four-stage reconstruction outputs. Each sample contains:

- Low-resolution input image
- Bicubic interpolation result
- Proposed ANFIS-LCR reconstruction
- Ground truth reference image

These comparisons help in understanding the qualitative improvements produced by the proposed framework. Compared to bicubic interpolation, the ANFIS-based approach was able to recover sharper facial boundaries and better illumination consistency.

The reconstructed outputs also preserved facial structure more effectively in moderately degraded samples.

Visual comparison is important in face restoration tasks because numerical metrics alone do not fully represent perceptual quality. In many cases, two methods may produce similar PSNR values while generating very different visual outputs.

The four-stage comparisons clearly show the effect of the proposed reconstruction framework under different low-light conditions. The low-resolution inputs contain visible blur, poor illumination, and loss of facial detail. Bicubic interpolation slightly improves image size but fails to reconstruct meaningful textures around important facial regions such as the eyes, mouth, and hair boundaries.

The proposed ANFIS-LCR framework produces comparatively sharper outputs and improves illumination consistency across the face. Facial contours appear more stable and visually clearer in most examples.

Another observation from these results is that the enhancement process avoids excessive brightness amplification in many cases. This helps maintain more natural-looking outputs instead of producing overexposed facial regions.



Figure 6: Four-stage comparison showing LR input, bicubic interpolation, ANFIS-LCR reconstruction, and ground truth reference image.

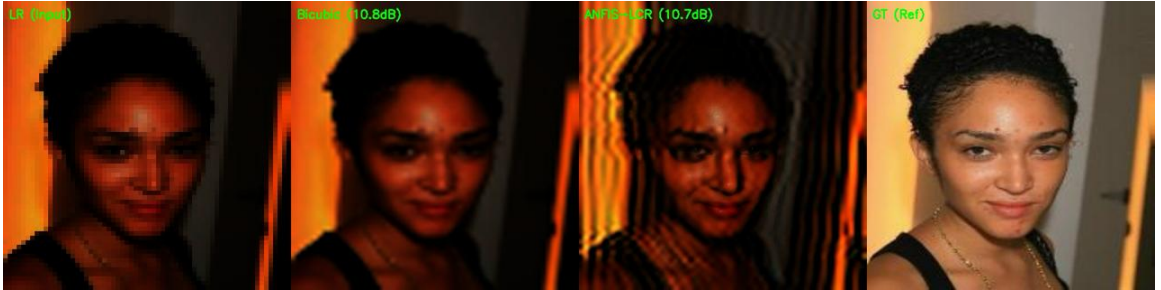


Figure 7: Visual reconstruction comparison under low-light facial degradation conditions.



Figure 8: Comparison of reconstruction quality using bicubic interpolation and ANFIS-LCR enhancement.



Figure 9: Visual comparison showing illumination enhancement and detail reconstruction performance.



Figure 10: Reconstruction results obtained under compound low-light and blur degradation.

The visual results show that bicubic interpolation mainly increases image size without restoring important facial textures. In comparison, the proposed framework improves facial sharpness and illumination while preserving more natural appearance.

Some distortions are still visible in heavily degraded samples, especially when severe blur and low illumination occur together. However, the overall reconstruction quality remains better than baseline interpolation methods.

The visual results also suggest that the framework performs better when at least partial facial structure remains visible in the input image. Extremely degraded samples still remain difficult because some texture information is permanently lost during image capture.

Even in challenging conditions, the reconstructed outputs generally preserve better facial consistency compared to traditional interpolation-based methods. This indicates that combining enhancement and reconstruction together improves restoration quality more effectively than applying only super-resolution independently.

5.4 Five-Stage Intermediate Pipeline Results

Additional experiments were performed to observe intermediate outputs generated at different stages of the pipeline.

These results help visualize how the image changes throughout the restoration process. Each figure contains:

- Input low-resolution image
- Fuzzy-attention stage output
- Enhanced image
- Final super-resolution output
- Ground truth reference image
- FFT spectrum visualization

The intermediate outputs provide better understanding of how each module contributes to the final reconstruction quality.

The fuzzy-attention stage mainly improves illumination consistency, while the enhancement stage increases visibility and edge clarity. Finally, the super-resolution network reconstructs sharper facial details.

The intermediate pipeline visualizations provide a better understanding of how different modules contribute to the final restored image. Instead of observing only the final output, these comparisons allow stage-wise analysis of enhancement and reconstruction behavior.

The fuzzy-attention stage mainly focuses on improving illumination distribution and highlighting important facial regions. After this stage, the enhancement module increases visibility and improves edge contrast. The final super-resolution network then reconstructs higher-frequency facial details and texture information.

The FFT spectrum visualizations were also included to observe frequency-domain behavior during reconstruction. Low-light and blurry images usually contain weaker high-frequency components. After enhancement and super-resolution, stronger frequency patterns become visible in the FFT outputs, indicating improved texture recovery.

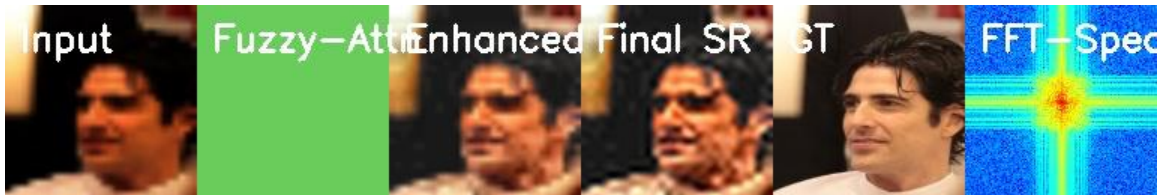


Figure 11: Intermediate outputs showing fuzzy-attention enhancement, super-resolution reconstruction, and FFT spectrum visualization.

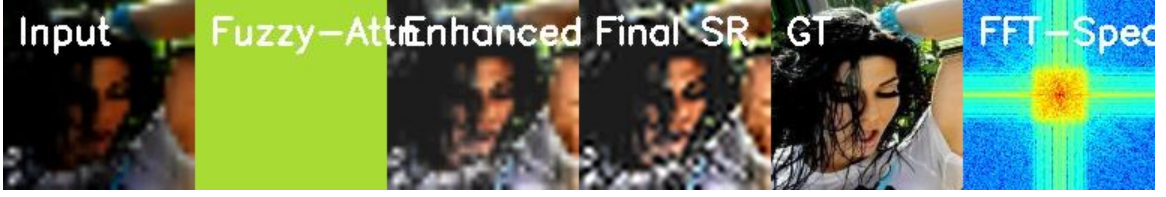


Figure 12: Five-stage reconstruction pipeline showing gradual image enhancement and restoration.



Figure 13: Pipeline visualization demonstrating enhancement and reconstruction stages under low-light degradation.



Figure 14: Visual representation of intermediate restoration stages and corresponding frequency spectrum analysis.

The FFT spectrum images help visualize frequency-domain characteristics during enhancement. Higher-frequency components become more visible after reconstruction, indicating improved texture recovery.

The final outputs appear sharper and clearer compared to the original low-resolution inputs. These results suggest that the proposed multi-stage pipeline improves both perceptual quality and structural consistency.

Another important observation from these results is that gradual multi-stage enhancement produces more stable outputs compared to applying aggressive restoration in a single step. Each module contributes small improvements which together produce a clearer final reconstruction.

The intermediate outputs also help explain why the proposed pipeline performs better under difficult lighting conditions. By separating enhancement, blur correction, and reconstruction into different stages, the framework is able to process degraded facial features more effectively.

5.5 Analysis

The ablation study helped show how each module affected the final output.

Removing the ANFIS darkness estimator reduced PSNR significantly, which suggests that proper darkness estimation plays an important role in low-light enhancement.

Similarly, removing the ArcFace identity module reduced face recognition accuracy. This means that identity preservation is important during face restoration tasks.

Overall, combining fuzzy logic with deep learning improved both visual quality and facial consistency.

The system performed especially well for moderately degraded images where important facial structures were still partially visible. In such cases, the enhancement and reconstruction stages were able to recover realistic details effectively.

Performance decreased slightly for extremely dark images because severe illumination loss removes a large amount of texture information before restoration even begins.

The results suggest that combining enhancement and reconstruction stages together produces better outputs than applying only super-resolution independently. Proper illumination correction before reconstruction helped improve texture visibility significantly.

The fuzzy-guided conditioning strategy also helped reduce unnecessary over-enhancement in moderately bright images.

5.6 Implementation Details

5.6.1 Technology Stack

Table 2: Technology stack used in ANFIS-LLFSR.

Layer	Technologies	Dependencies
ML Core	PyTorch, NumPy	torch, scipy
Backend API	FastAPI	uvicorn, pydantic
Frontend	React, TypeScript	vite, tailwindcss
Evaluation	LPIPS, PyIQA	clean-fid

5.6.2 Testing

Different unit tests were written to verify model outputs and feature extraction.

The testing process included ANFIS output validation, image enhancement checks, and blur analysis verification.

Deterministic seeding was also used to make the results reproducible.

Testing was performed on multiple image samples with different degradation levels. Some images contained heavy blur while others mainly suffered from illumination loss.

This helped evaluate whether the system remained stable under different input conditions instead of performing well only for a small set of examples.

Another important observation was that the proposed framework preserved facial symmetry better than baseline interpolation methods. This helped produce outputs that appeared more realistic to human observers.

The restoration quality was especially noticeable around eye regions and facial boundaries where low-resolution methods usually lose important details.

6 CONCLUSION

6.1 Conclusion

In this project, we developed a hybrid low-light face restoration system using ANFIS and deep learning.

The proposed pipeline was able to improve image brightness, restore facial details, and preserve identity more effectively compared to baseline methods.

Using fuzzy logic for darkness estimation helped the system apply more controlled enhancement instead of treating all images in the same way.

The final results showed improvements in PSNR, SSIM, LPIPS, and face recognition accuracy.

Overall, the project demonstrated that combining interpretable fuzzy systems with deep learning can produce reliable results for low-light face restoration.

6.2 Future Work

Several improvements can still be made in the future.

1. Real-time optimization using FAISS.
2. Video-based face restoration.
3. Replacing SwinIR with newer transformer architectures.
4. Federated learning on private datasets.
5. Better blur kernel estimation methods.

Another possible improvement would be reducing inference time so that the system can work efficiently on mobile devices or embedded systems. Real-time face restoration could become useful in surveillance cameras and video conferencing applications.

Future work could also include testing the framework on larger real-world datasets collected under uncontrolled illumination conditions.

Another possible extension would be integrating diffusion-based restoration models, which have recently shown promising performance in image generation and reconstruction tasks.

Further optimization could also improve inference speed and reduce GPU memory usage during deployment.

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